Digital video forensic report

Case SIH26150-DEMO · examiner Team SPARCX · generated 2026-10-04 11:42:05 UTC · dvrf 0.1.0

1. Summary

Recorder format family	hikvision (confidence 0.99) — Hikvision, HiWatch, Prama Hikvision, Hikvision OEMs (verify per model)
Analysis mode	native
Video frames recovered	5401 in 6 clip(s) across 3 camera(s)
By evidence level	index-confirmed 1200 content-only 11 carved + timestamp 4190
Findings	3 critical · 2 warning · 1 info
Evidence integrity	Source hash identical before and after analysis; custody ledger: 6 entries, chain intact

Evidence levels. *Index-confirmed*: listed in the recorder's own live index. *Carved + timestamp*: recovered from disk with a timestamp the recorder wrote into the video container. *Content-only*: recovered video whose camera and time come from analysing the picture (burned-in overlay) or are unknown. Machine-generated triage (section 6) is not a finding of fact.

2. Evidence and acquisition

Source	`/tmp/k4a/images/hikvision_overwritten.img`
Size	352,321,536 bytes
MD5 (before analysis)	`ea476dd4500bd94ea418926241318b96`
SHA-256 (before analysis)	`1b6e3fc590fd7f7214a3f4d2810fabb980ba453a1980a09f9f18e8eef4aadb20`
SHA-256 (after analysis)	`1b6e3fc590fd7f7214a3f4d2810fabb980ba453a1980a09f9f18e8eef4aadb20`

3. Device identification

Candidate family	Confidence	Basis
hikvision (hikvision)	0.99	`{'signature': 'HIKVISION@HANGZHOU', 'master_sector_offset': 512, 'partition_offset': 0}`
unknown (universal)	0.3	`{'ps_pack_headers_first_256MiB': 5401}`
dahua-dhav (dahua)	0.0	`{}`

Filesystem metadata

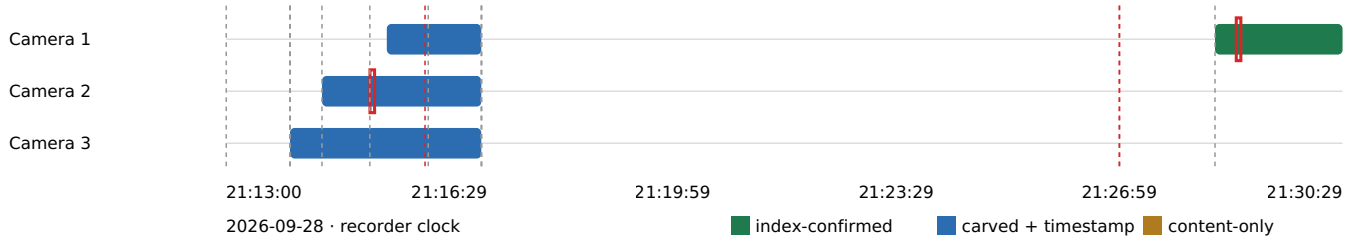
offset	`512`
partition_offset	`0`
capacity	`352321536`
logs_offset	`1048576`
logs_size	`1048576`
video_offset	`4194304`

block_size	4194304
total_blocks	83
btree1_offset	2097152
btree1_size	524288
btree2_offset	2621440
btree2_size	524288
init_time	1790630820

4. Findings

Severity	Type	Finding
critical	format-event	System log records a disk initialise/format at 2026-09-28T21:27:00+00:00
critical	index-reset	4 data block(s) hold video that the live index does not list; 4190 recovered frames predate the filesystem initialisation time 2026-09-28T21:27:00+00:00
critical	clock-change	Camera 1: overlay-to-recorder offset changes during recording (+0 s to -900 s) between 21:15:58 and 21:16:07 recorder time: possible clock adjustment
warning	clock-drift	Camera 2: burned-in clock is +47 s from the recorder timestamp
warning	clock-drift	Camera 3: burned-in clock is -312 s from the recorder timestamp
info	recording-gap	Camera 1: no recovered video for 690 s between C0001-CH01 and C0002-CH01

5. Timeline



Bars: recovered clips coloured by evidence level. Red outlines: motion triage. Dashed lines: recorder log entries (red = format/initialise).

Cameras and clock drift

Channel	Burned-in label	Clips	Frames	Overlay clock vs recorder
1	CAM-01 MAIN GATE	2	2090	+0 s (18/24 readings agree)
2	CAM-02 PARKING	1	1500	+47 s (24/24 readings agree)
3	CAM-03 CORRIDOR	1	1800	-312 s (56/59 readings agree)

Recovered clips

Clip	Ch	Start (recorder clock)	End	Frames	Evidence	Time source	Exported file and SHA-256
C0001-CH01	1	2026-09-28 21:15:31.0	2026-09-28 21:16:59.9	890	carved + timestamp	ofni	C0001-CH01.mp4 d387c723d4c139853e48ad6e1aea9abafeab2a4f992cd56085940ebbce0735d2
C0002-CH01	1	2026-09-28 21:28:30.0	2026-09-28 21:30:29.9	1200	index-confirmed	index	C0002-CH01.mp4 9500bbd90ad43301eb98eaed4fa2479904e667bbb8ebb487956b8ad624b2dcc3
C0003-CH02	2	2026-09-28 21:14:30.0	2026-09-28 21:16:59.9	1500	carved + timestamp	ofni	C0003-CH02.mp4 6f78181db51fcd287714b94e075f153c0d8a5e38473ac2bb500f8bf76742c28f
C0004-CH03	3	2026-09-28 21:14:00.0	2026-09-28 21:16:59.9	1800	carved + timestamp	ofni	C0004-CH03.mp4 872b5ede83cc55289045e16c62f841c7b0a4779bdded43db620256a4ade6f07e
C0005	?	—	—	2	content-only	none	C0005.h264 1b21f3e38463e23f76171c46fa90d4f68024752020801e5042685a815acb023e
C0006	?	—	—	9	content-only	none	C0006.h264 9b93fbdc87c26829e2a45749ae716c06a9f905aa619ca72fd7d2461a9075cb30

6. Machine triage (not findings)

> Automatic detections are leads for a human examiner. Each states its method and confidence. None is evidence by itself and none appears in the certificate.

Channel	From	To	Type	Confidence	Basis
1	2026-09-28 21:28:50.0	2026-09-28 21:28:53.7	moving-object	0.86	background subtraction, compact foreground blob
2	2026-09-28 21:15:15.2	2026-09-28 21:15:18.7	moving-object	0.79	background subtraction, compact foreground blob

7. Recorder system log

Time (recorder clock)	Type	Entry
2026-09-28 21:13:00.0	power/login/shutdown	Power On
2026-09-28 21:14:00.0	hdd/recording	Start Recording CH01
2026-09-28 21:14:00.0	hdd/recording	Start Recording CH02
2026-09-28 21:14:00.0	hdd/recording	Start Recording CH03
2026-09-28 21:14:30.0	motion	Start Motion Detect CH01
2026-09-28 21:15:15.0	motion	Start Motion Detect CH02
2026-09-28 21:16:10.0	motion	Start Motion Detect CH03

Time (recorder clock)	Type	Entry
2026-09-28 21:17:00.0	hdd/recording	Stop Recording CH01
2026-09-28 21:17:00.0	hdd/recording	Stop Recording CH02
2026-09-28 21:17:00.0	hdd/recording	Stop Recording CH03
2026-09-28 21:27:00.0	hdd/recording	HDD Initialize
2026-09-28 21:28:30.0	hdd/recording	Start Recording CH01

8. Method, tools and limitations

- Recovery walks the whole source. A native parser for the detected family runs first; a vendor-agnostic H.264 carver then sweeps every byte the parser did not claim.
- Frames are validated structurally (Dahua: matching DHAV header and footer; MPEG-PS: complete pack and PES lengths) before they are counted.
- Times are shown as stored by the recorder (its local clock); overlay clocks are compared with them per camera.
- Tools: ffmpeg 6.1.1-3ubuntu5; tesseract 5.3.4; opencv 4.13.0.
- Dahua DHFS index structures are not parsed; Dahua-family frames are carved and carry the recorder timestamp from each DHAV header (evidence level: carved + timestamp).
- The universal path attributes camera and time from burned-in overlays; footage without overlays is recovered as content-only. PRNU sensor-noise clustering is not yet implemented.
- Motion triage misses objects against full-frame moving texture (heavy rain, foliage, noise).

Annex A. Data for the BSA s.63(4) certificate

> This sheet collects the facts the Schedule certificate asks for. Transcribe them onto the official Schedule form (Part A by the party producing the record; Part B by the expert). It is not itself the certificate. Fields marked ✎ must be completed by a person.

Part A — produced electronic record

Device type	Digital video recorder (DVR/NVR) storage
Make / model / serial number	✎ (detected format family: hikvision)
Storage medium details	✎ HDD make, model, serial, capacity (352,321,536 bytes read)
Under lawful control of	✎
Hash algorithm(s)	SHA-256 (primary), MD5 (secondary)
Hash of source / forensic image	`SHA-256 1b6e3fc590fd7f7214a3f4d2810fabb980ba453a1980a09f9f18e8eef4aadb20` `MD5 ea476dd4500bd94ea418926241318b96`

Produced records (clips) and their SHA-256	`C0001-CH01.mp4 d387c723d4c139853e48ad6e1aea9abafeab2a4f992cd56085940ebbce0735d2`
	`C0002-CH01.mp4 9500bbd90ad43301eb98eaed4fa2479904e667bbb8ebb487956b8ad624b2dcc3`
	`C0003-CH02.mp4 6f78181db51fcd287714b94e075f153c0d8a5e38473ac2bb500f8bf76742c28f`
	`C0004-CH03.mp4 872b5ede83cc55289045e16c62f841c7b0a4779bdded43db620256a4ade6f07e`

Part B — expert

Examiner	Team SPARCX ✎ designation and organisation
Examination performed	2026-10-04 11:42:05 UTC using dvrf 0.1.0
Method summary	Read-only analysis of the hashed source; format-family identification; index parsing and structural carving; per-clip export with SHA-256; custody ledger with hash chain.
Integrity statement data	Source SHA-256 before analysis equals after analysis.
Signature, date, place	✎

Annex B. Chain of custody

Ledger status: **6 entries, chain intact**

#	UTC	Actor	Action	Entry hash
0	2026-10-04T11:41:09+00:00	root	case-opened	`b794702da2f82efe3bf11940…`
1	2026-10-04T11:41:10+00:00	root	evidence-registered	`fd53fcb13a7b19b9a7ba8ad5…`
2	2026-10-04T11:41:12+00:00	root	recovery	`9eeaadf4aa029ad0b5f86f55…`
3	2026-10-04T11:41:54+00:00	root	clips-exported	`c5440ef46b0c64328a6c272a…`
4	2026-10-04T11:42:05+00:00	root	post-analysis-verification	`3ab019280ab63111c823f12d…`
5	2026-10-04T11:42:05+00:00	root	results-written	`be5c8b7a00af2fbcc80b946b…`